



University of San Carlos | Department of
COMPUTER ENGINEERING

CpE 3104 – Microprocessors

8086 μ P Hardware Specifications

Unit 5: MEMORY INTERFACING

© 2025

Intel 8086 μ P

- First 16-bit microprocessor released by Intel in the year 1978
- Manufactured using **high-density metal oxide semiconductor (HMOS)** III technique
- Approximately made up of 29,000 transistors
- 40-pin dual-in-line package (DIP)
- +5V Power Supply
- No internal clock, uses **external asymmetric clock source** with **33% duty cycle**
- **20-bit address** to access **memory**; can **address up to $2^{20} = 1\text{Megabyte (1MB)}$** of memory space

Intel 8086 μ P

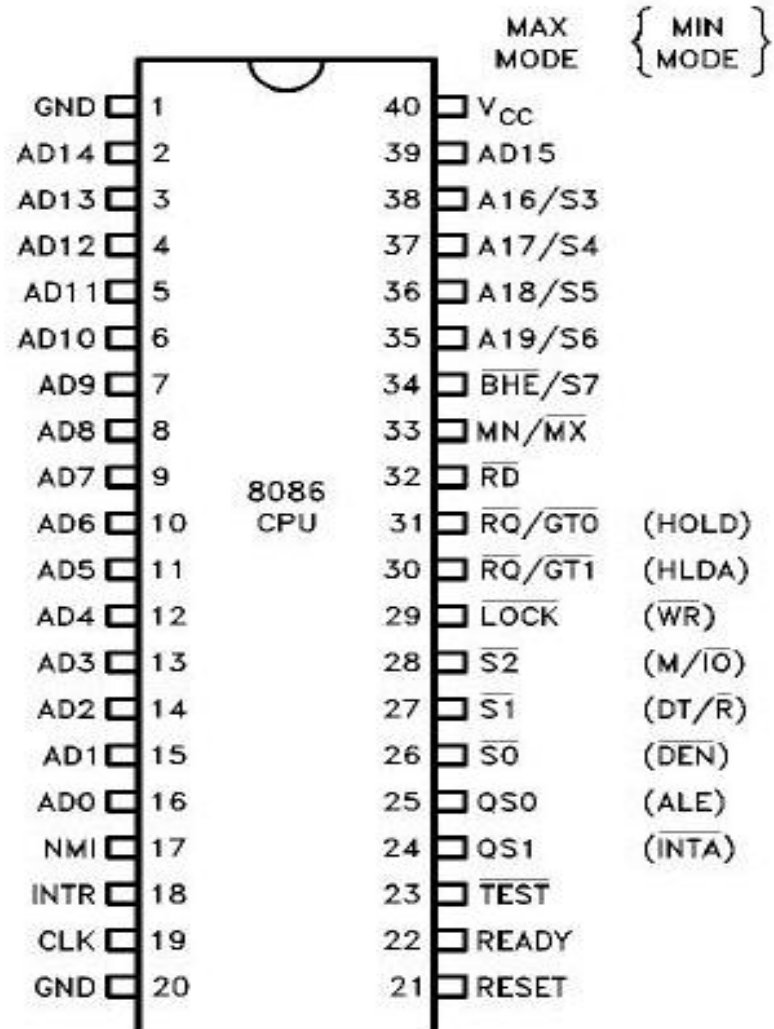
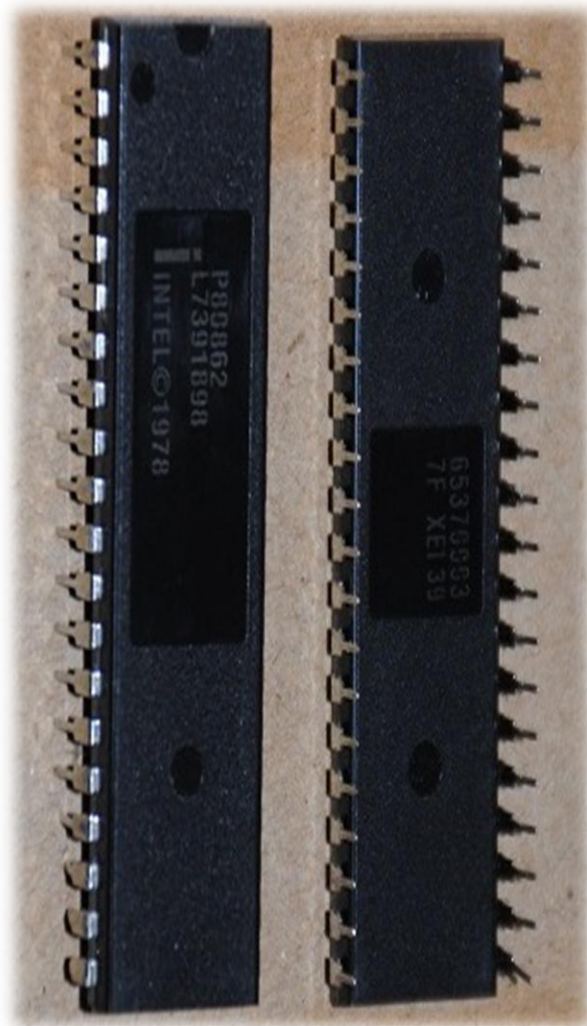
- **Addressable memory space** is organized into **two banks** of **512KB** each: **Even** (or **lower**) bank and **Odd** (or **higher**) bank
- Address line **A_0** is used to select **even bank** and control signal **BHE'** is used to access **odd bank**
- Uses a separate **16-bit address** for *I/O mapped devices*; can generate **$2^{16} = 64K$** addresses
- Operates in **two modes**: *minimum mode* and *maximum mode*; decided by the signal at **MN** and **MX'** pins
- Has **16 data lines** ($D_0 - D_{15}$)
- Uses time multiplexing for the Address, Data, and some Status lines



8086 μ P

Pin Configuration & Functions

8086 μ P Pin Configuration

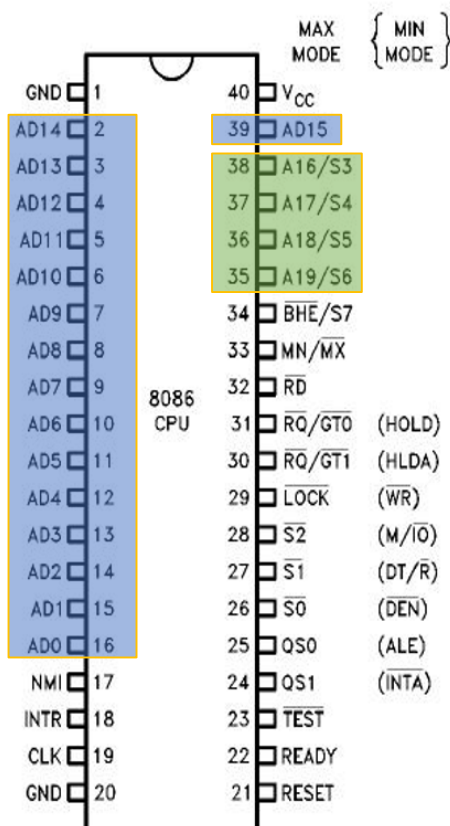


8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Common Signals



AD₀-AD₁₅ (Bidirectional)

Address/Data bus

- Low-order address bus; these are multiplexed with data.
- When AD lines are used to transmit memory address the symbol A is used instead of AD, for example A₀-A₁₅.
- When data are transmitted over AD lines the symbol D is used in place of AD, for example D₀-D₇, D₈-D₁₅ or D₀-D₁₅.

A₁₆/S₃, A₁₇/S₄, A₁₈/S₅, A₁₉/S₆

Address/Status

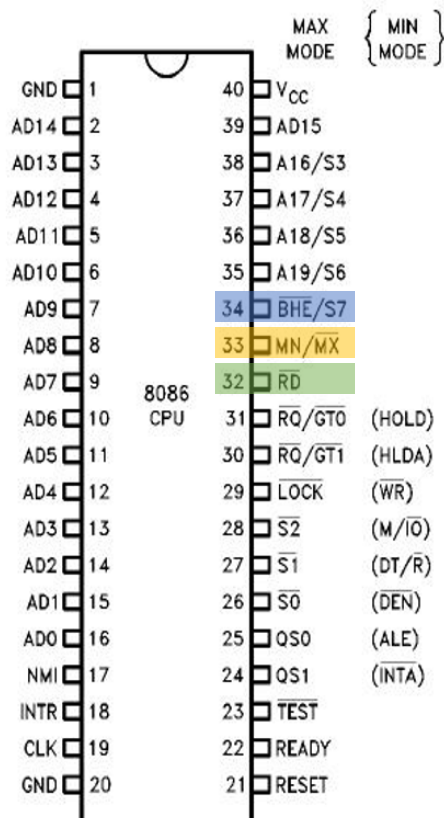
- High-order address bus.
- These are multiplexed with status signals.

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Common Signals



BHE (Active Low)/S₇ (Output) Bus High Enable/Status

- It is used to enable data onto the most significant half of data bus, D₈-D₁₅. An 8-bit device connected to upper half of the data bus use BHE (Active Low) signal. It is multiplexed with status signal S₇.

MN/ MX (Active Low) Minimum /Maximum

- This pin signal indicates what mode the processor is to operate in.

RD (Read) (Active Low) Read

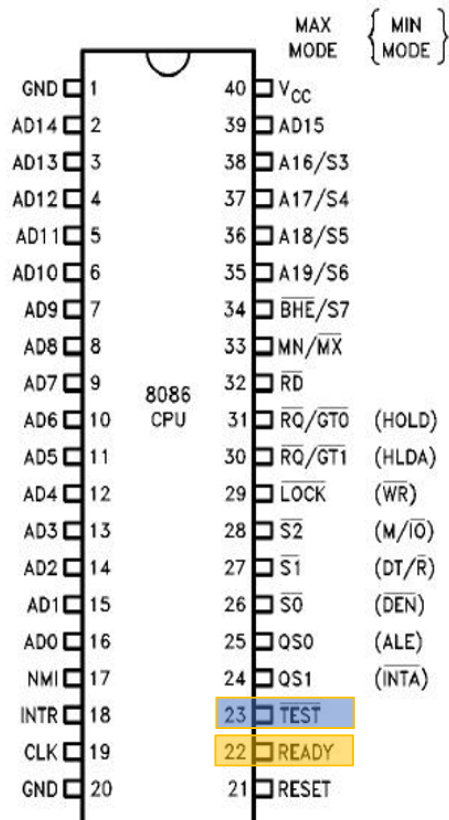
- The signal is used for read operation.
- It is an output signal.
- It is active when low.

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Common Signals



TEST

- Test is an active low input and is tested by the 'WAIT' instruction.
- 8086 μ P will enter into a wait state after execution of the WAIT instruction and will resume execution only when the TEST is made low by an active hardware.
- This is used to synchronize an external activity to the processor internal operation.

READY

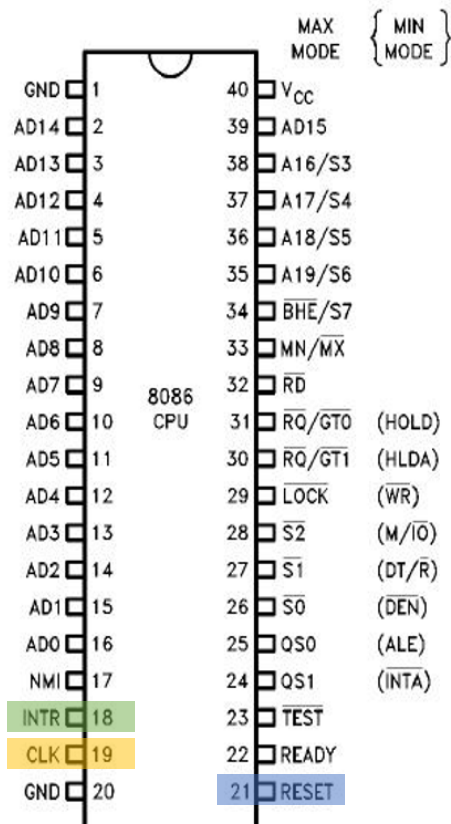
- This is the acknowledgement from the slow device or memory that they have completed the data transfer.
- The signal made available by the devices is synchronized by the 8284A clock generator to provide ready input to the 8086 microprocessor.
- The signal is active high.

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Common Signals



RESET (Input)

- It causes the processor to immediately terminate its present activity.
- The signal must be active HIGH for at least four clock cycles.

CLK (Clock)

- The clock input provides the basic timing for processor operation and bus control activity. It's an asymmetric square wave with 33% duty cycle.

INTR (Interrupt Request)

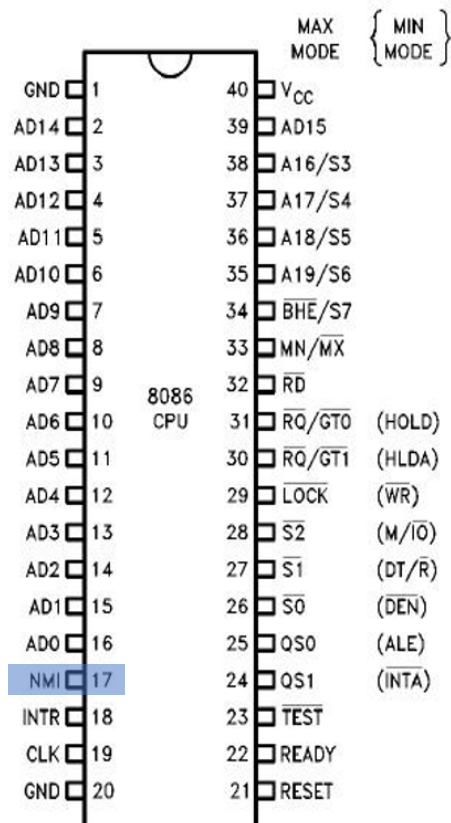
- This is a triggered input. This is sampled during the last clock cycles of each instruction to determine the availability of the request. If any interrupt request is pending, the processor enters the interrupt acknowledge cycle.
- This signal is active high and internally synchronized.

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Common Signals



NMI (Non-Maskable Interrupt)

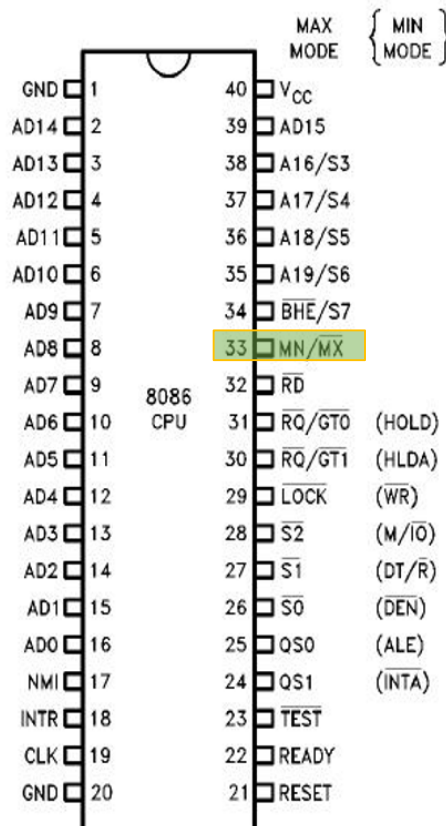
- This input is similar to INTR except that the NMI interrupt does not check to see whether the IF flag bit is a logic 1.
- This interrupt input uses interrupt vector 2 if it is activated.

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Min/Max Pins



• The 8086 microprocessor can work in two modes of operations : *Minimum mode* and *Maximum mode*.

• In the Minimum mode of operation, the microprocessor do not associate with any co-processors and cannot be used for multiprocessor systems.

• In the Maximum mode, the 8086 μ P can work in multiprocessor or co-processor configuration.

• Minimum or Maximum mode operations are decided by the pin MN/ $\overline{MX}$ (active low).

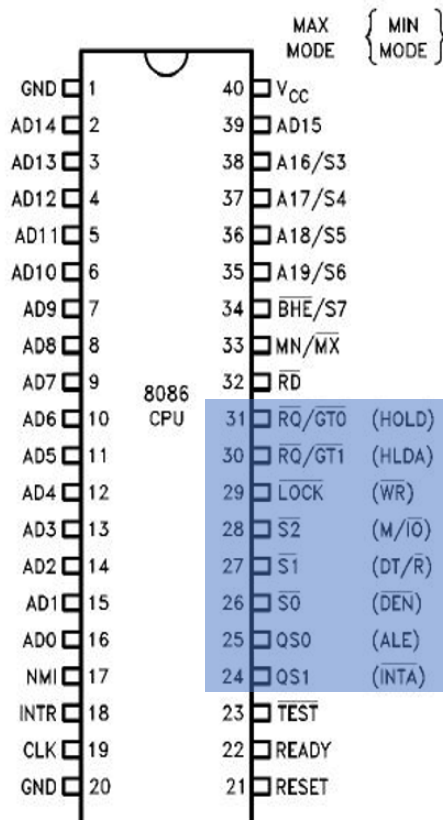
• When this *pin 33* is high, 8086 μ P operates in Minimum mode, otherwise, it operates in Maximum mode.

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Minimum Mode Signals



Pins 24 to 31

- For minimum mode operation, the $MN/\overline{MX}$ is tied to V_{CC} (logic high).
- 8086 μ P itself generates all the bus control signals.

DT/ $\overline{R}$ (Data Transmit/Receive): Output signal from the processor to control the direction of data flow through the data transceivers (1 = transmitting; 0 = receiving)

$\overline{DEN}$ (Data Enable/Data Bus Enable): Output signal from the processor used as output enable for the transceivers

ALE (Address Latch Enable): Used to demultiplex the address and data lines using external latches

M/ $\overline{IO}$ (Memory/IO): Used to differentiate memory access and I/O access. For memory reference instructions, it is **high**. For IN and OUT instructions, it is **low**.

$\overline{WR}$ (Write control signal): asserted **low** whenever processor writes data to memory or I/O port

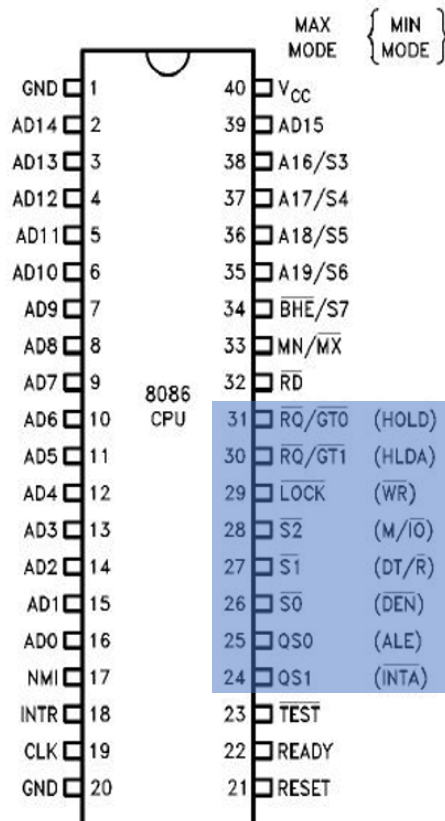
$\overline{INTA}$ (Interrupt Acknowledge): When the interrupt request is accepted by the processor, the output is **low** on this line.

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Minimum Mode Signals



Pins 24 to 31

- For minimum mode operation, the $\overline{MN}/\overline{MX}$ is tied to V_{CC} (logic high).
- 8086 μ P itself generates all the bus control signals.

HOLD Input signal to the processor that informs the bus master as a request to grant the control of the bus.

Usually used by the DMA controller to get the control of the bus.

HLDA (Hold Acknowledge) Acknowledge signal by the processor to the bus master requesting the control of the bus through HOLD.

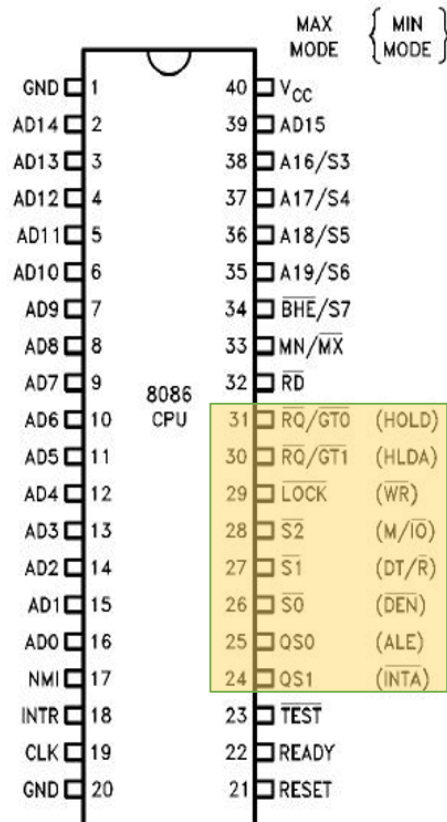
The acknowledge is asserted high, when the processor accepts HOLD.

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Maximum Mode Signals



Pins 24 to 31 are reassigned

- During maximum mode operation, the $MN/\overline{MX}$ is grounded (logic low).

$\overline{S_0}, \overline{S_1}, \overline{S_2}$

Status signals; used by the 8086 bus controller to generate bus timing and control signals. These are decoded as shown.

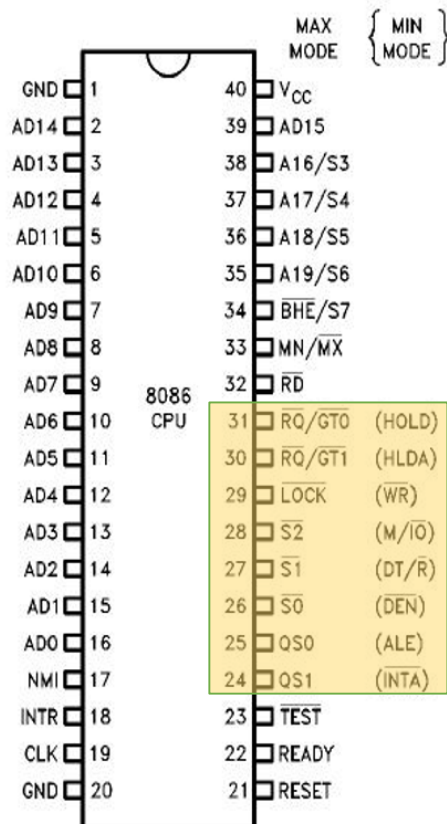
Status Signal			Machine Cycle
$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$	
0	0	0	Interrupt acknowledge
0	0	1	Read I/O port
0	1	0	Write I/O port
0	1	1	Halt
1	0	0	Code access
1	0	1	Read memory
1	1	0	Write memory
1	1	1	Passive/Inactive

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Maximum Mode Signals



Pins 24 to 31 are reassigned

- During maximum mode operation, the $\overline{MN}/\overline{MX}$ is grounded (logic low).

QS₀, QS₁ (Queue Status) The processor provides the status of queue in these lines.

The queue status can be used by external device to track the internal status of the queue in 8086.

The output on QS₀ and QS₁ can be interpreted as shown in the table.

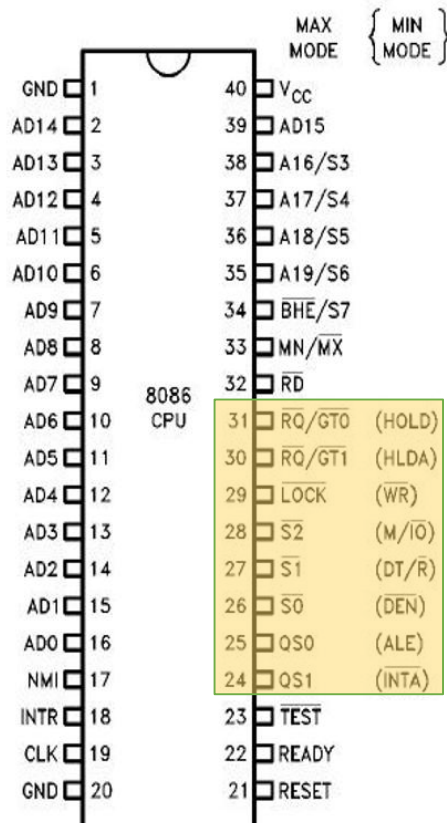
Queue status		Queue operation
QS ₁	QS ₀	
0	0	No operation
0	1	First byte of an opcode from queue
1	0	Empty the queue
1	1	Subsequent byte from queue

8086 μ P Pin Functions

8086 Microprocessor

Pins and Signals

Maximum Mode Signals



Pins 24 to 31 are reassigned

- During maximum mode operation, the $MN/\overline{MX}$ is grounded (logic low).

$\overline{RQ}/\overline{GT_0}$,
 $\overline{RQ}/\overline{GT_1}$

(Request/Grant) These requests are used by other local bus masters to force processor to release the local bus at the end of the processor's current bus cycle.

These pins are bidirectional.

The request on GT_0 will have higher priority than GT_1 .

$\overline{LOCK}$

An output signal activated by the LOCK prefix instruction.

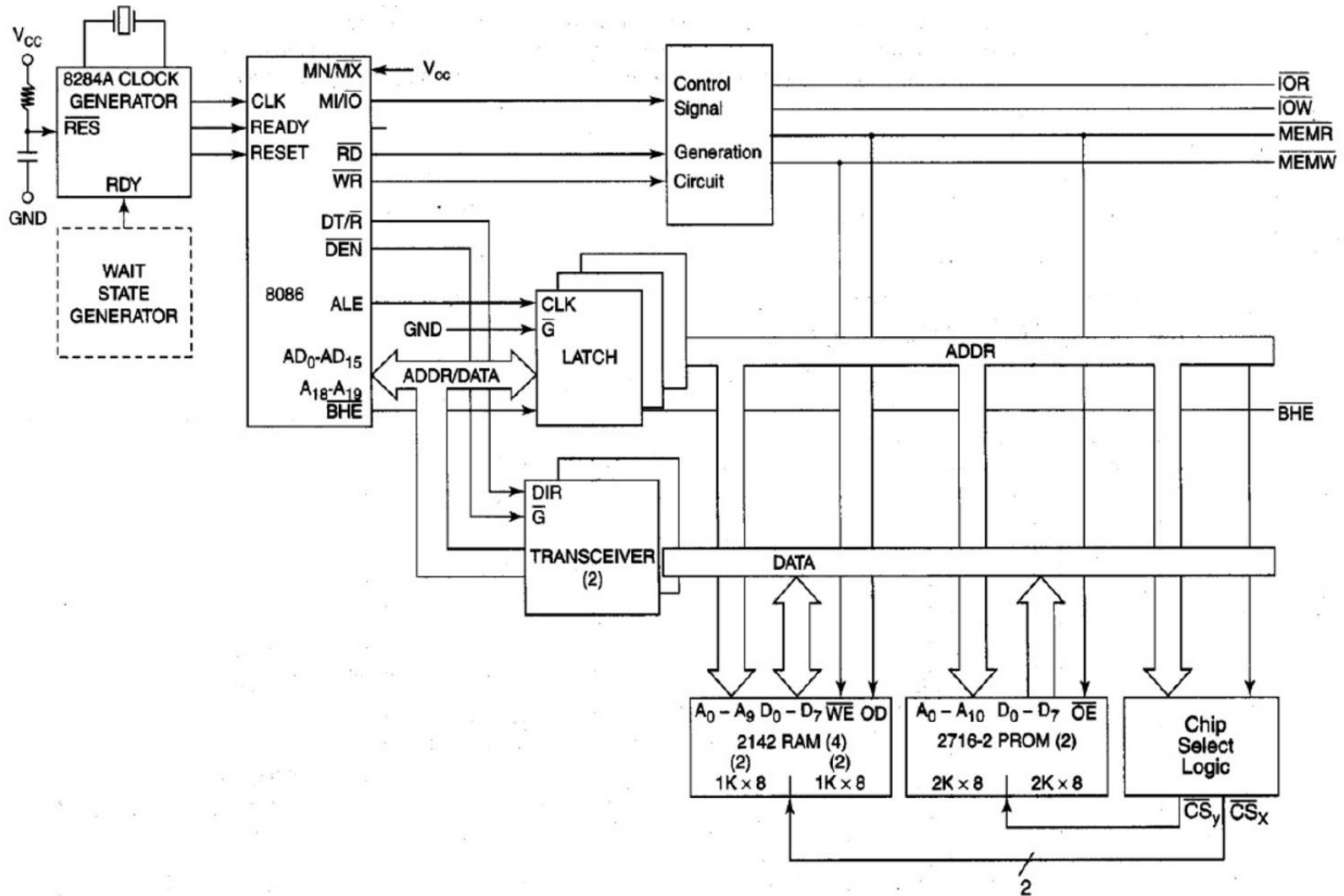
Remains active until the completion of the instruction prefixed by LOCK.

The 8086 microprocessor outputs low on the LOCK pin while executing an instruction prefixed by LOCK to prevent other bus masters from gaining control of the system bus.

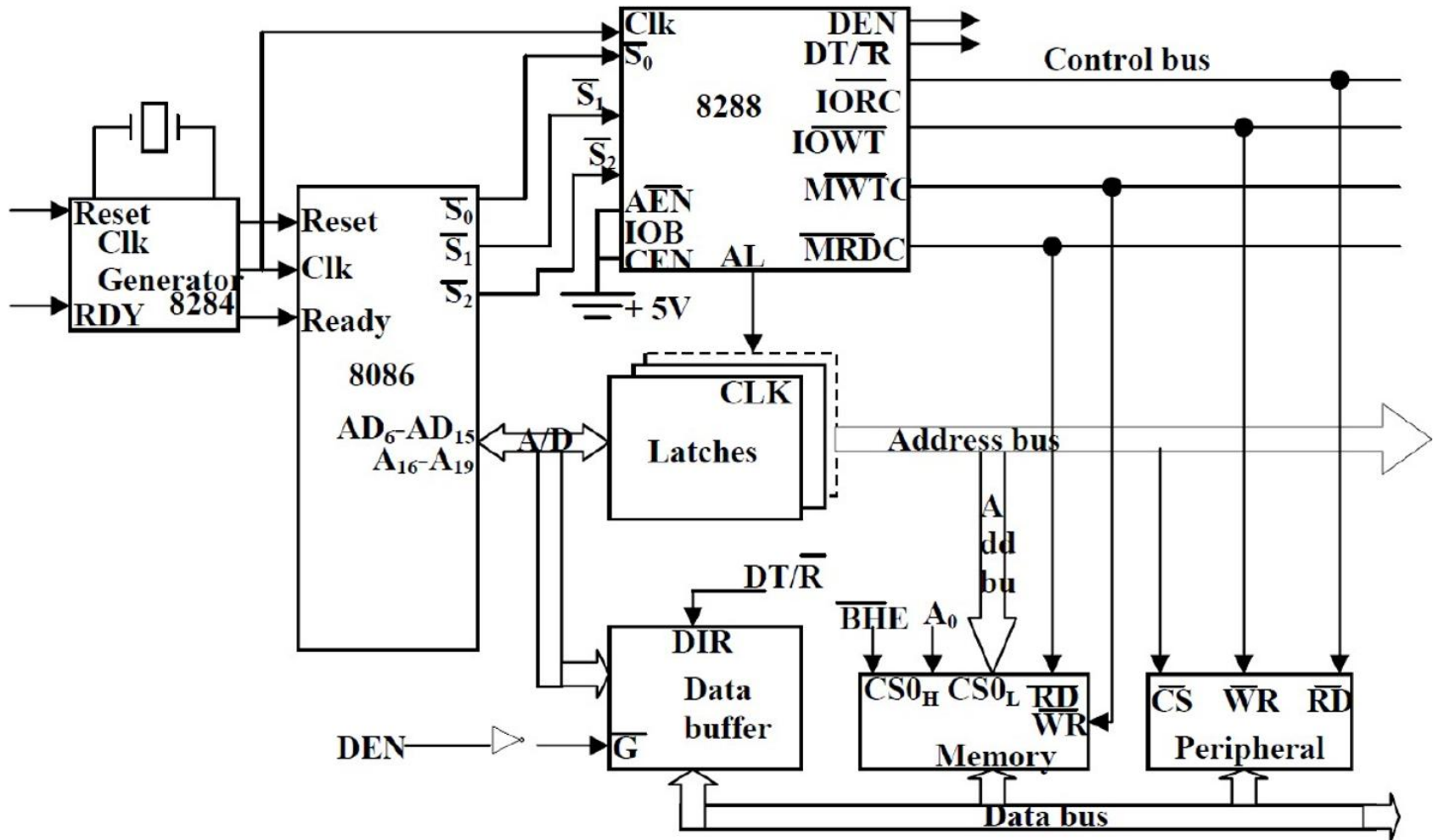


8086 μ P Interface Block Diagram

8086 μ P Minimum Mode Interface Block Diagram



8086 μ P Maximum Mode Interface Block Diagram





University of San Carlos | Department of
COMPUTER ENGINEERING

End of Lecture